GAME SENSE

ESPORTS PERFORMANCE & GAMING HEALTH ANALYTICS | KEY INSIGHTS

Project Summary

The project analyzes 250,000 esports player-session records using Python and data visualization, with the results presented in a Power BI dashboard. The findings below are reported in the project documentation and describe observed patterns rather than causal effects.

1. Sleep Duration and Win Rate

Sleep Group	Win Rate
Less than 5 hours	47.95%
5–7 hours	50.71%
7–9 hours	51.61%
9+ hours	52.12%

Insight: The 9+ hour sleep group had the highest observed win rate (52.12%), while the under-5-hour group had the lowest (47.95%). The documentation also reports lower average reaction time and higher average HRV among groups with more sleep.

2. Reaction Time and Win Rate

Reaction Time Group	Win Rate
Less than 220 ms	50.19%
220–250 ms	54.20%
250–280 ms	50.36%
280–310 ms	46.93%
310+ ms	44.89%

Insight: The 220–250 ms group had the highest observed win rate (54.20%). Win rate declined across the slower reaction-time groups, reaching 44.89% for the 310+ ms group.

3. APM and Match Outcomes

APM Group	Win Rate
Less than 100	63.64%
100–200	44.91%
200–300	49.88%
300–400	51.71%
400+	55.34%

Insight: The under-100 APM group had the highest observed win rate (63.64%). This result suggests that APM alone is not a reliable standalone explanation of match success.

4. Energy Drinks and Heart Rate

The reported average heart rate increased from approximately 80 BPM for players consuming zero energy drinks to approximately 97 BPM for players consuming four. The analysis did not show a clear corresponding improvement in win rate.

5. Game Genre Comparison

Game Genre	Win Rate	Average HRV
Fighting	50.63%	87
MOBA	49.74%	87
FPS	46.46%	88
Battle Royale	45.61%	88

Insight: Fighting had the highest observed win rate (50.63%) and Battle Royale the lowest (45.61%). Average HRV scores were similar across the four genres, ranging from 87 to 88.

6. Blue-Light Filter Usage

Approximately 65% of players used a blue-light filter in the dataset, indicating that this was a common behavior among the analyzed players.

Overall Conclusion

The documented analysis found that sleep duration and reaction-time groups showed clearer differences in win rate than a simple assumption that higher APM always means better outcomes. Energy drink consumption was associated with higher average heart rate, while average HRV varied little across the game genres studied.

Interpretation Caution

These findings are descriptive associations from the dataset. They do not prove that sleep duration, reaction time, energy drinks, or game genre directly cause match outcomes. Other variables may influence the observed patterns.